

venerdì 17 aprile 2026 14:51

25c QAM

$$P_{E3} = 10^{-8}$$

$$S_{min} \propto [w]$$

$$P_b: \frac{2}{9} \cdot \frac{2^{-\frac{\gamma}{21,25}}}{3\pi}, \quad \gamma: \frac{65}{10} \Rightarrow \gamma = 329$$

$$\frac{S}{N} = \frac{E_b / T_b}{n_{0B}} = \frac{E_b f_b}{n_{0f_b}} = \frac{E_b}{n_0} \cdot f = 329 \cdot f = 2032$$

2) Cancel bino's anti-pole $\pm$ site

$$BCU, m=255, t=3$$

Q5: ?

3. Ziele

$$h = 8, r = 3 \Rightarrow 24$$

$$\{5 = 10^{-3} = \frac{e^{-\gamma}}{3\pi} \rightarrow \gamma = 4.66$$

$$\delta' = \delta \cdot \frac{K}{n} = 4,66 \cdot \frac{231}{255} = 4,66 \cdot 0,9 = 4,22$$

$$k_1' = \frac{2 \cdot 84,72}{39} = 1,6 \cdot 10^{-3}$$

$$P_{ep} = 1 - P_p = \sum_{k=0}^3 \binom{255}{k} (1,6 \cdot 10^{-3})^k (1 - 1,6 \cdot 10^{-3})^{255-k} = 8,2 \cdot 10^{-4}$$

$$P_{bc} = \frac{d}{m} \cdot 8,2 \cdot 10^{-4} = 2,2 \cdot 10^{-5}$$

3) 64 QAM, $P_s = 10^{-2}$, a perito di benzola,
ES un collega a simboli;

$$P'_f = ?$$

Soluzione

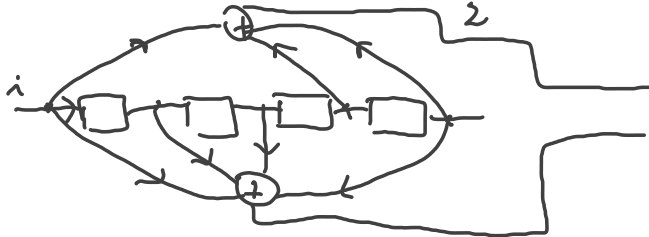
$C_4 = 2^h \rightarrow h = 6 \Rightarrow$ 10 byte è fatto da 6 bit

dato da due campi 4 mmi $\rightarrow$ 8 bit di controllo

$$P_{ep} : 1 - P_b = \sum_{k=1}^6 \binom{6}{k} (P_{simb})^k (1 - P_{simb})^{6-k} = 4,3 \cdot 10^{-4}$$

$$P'_{simb} = \frac{2}{63} \cdot 4,3 \cdot 10^{-4}$$

4) convoluzionale con $r = \frac{1}{2}$ $L = 5$



unioni per la probabilità?

ovvero la distanza tra parole a distanza minima

Soluzione

000...0 } tutte le parole di distanza 1
100...0

Simbolo tutti i bit $\Rightarrow$ 100...0 $\Rightarrow$ 11 01 01 10 11 ... 00

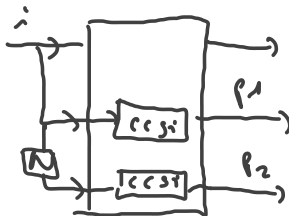
energia circa di 7 (7 "1") $\Rightarrow E = \frac{7}{2}$

$$P_{parit} = \frac{e^{-\frac{7}{2}}}{3\pi} ; P_{bparit} = \frac{1}{2} \frac{e^{-\frac{7}{2}}}{3\pi}$$

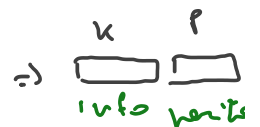
Se anche $P_{eb} = 10^{-2} \Rightarrow \gamma = 2,36 \left(\frac{e^{-\frac{7}{2}}}{3\pi} = 10^{-2} \right)$

$$P_{cod} = \frac{1}{2} \frac{e^{-2,36 \frac{7}{2}}}{3\pi} = 1,4 \cdot 10^{-5}$$

5) Turboencoder $r = \frac{1}{3}$



CCSI: codici convoluzionali
sistematici iterativi



$N = 65 \cdot 10^3$ 001000100 **ordine**

ordine = 6 $\cdot P_p = ?$

Soluzione

$$P_p = \frac{3}{N} \frac{e^{-\frac{7 \cdot \text{ordine}}{3}}}{3\pi} ; \gamma = 1,25 \Rightarrow 1,25 \text{ ordine} \rightarrow$$

$$\Rightarrow \frac{3,2 e^{-\frac{7 \cdot 25 \cdot 6}{3}}}{65000 \cdot 3\pi} = 7,5 \cdot 10^{-6}$$